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Aerodynamics ME 564

Aircraft Design Project

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1.0 Introduction

Every year in Canada, anywhere between 70 000 and 80 000 people are reported missing [1]. Many of these cases involve individuals who get lost, wander off or get stuck somewhere amongst the vast Canadian wilderness. Canada has the world's second largest wilderness as 90% of Canada's roughly 40 million population resides only 150 miles from the US border [2].

This results in an immense physical area which Canadian authorities are often tasked with searching for missing persons. The purpose of this report is thus to design an aircraft which can aide these authorities in their search and rescue operations for lost individuals. The aircraft will feature no crew or passengers and will be controlled completely remotely. The aircraft's design is to be based on existing, proven technologies to ensure reliability and simplicity.

This report will explore the complete design process of the aircraft, from the initial conceptual design objectives and requirements to the final configuration and performance characteristics.

2.0 Design Specifications

2.1 Conceptual Design

The initial concept for the aircraft was a remote-controlled drone which would search a region of interest for any signs of missing persons. The aircraft would include all necessary onboard avionics as well as a suite of onboard sensors and cameras. The primary functions of the aircraft would thus be remotely controlled directly by a human pilot in their search for lost individuals or by autonomously searching a specified area by performing predetermined sweeping operations.

Functionally, the design would be similar to that of military drones however with key differences in payload and construction. While military drones also include similar onboard avionics and sensor/camera suite, military drones do not internally carry the survival equipment to be dropped by the aircraft if survivors are located. Instead, they generally carry various externally mounted munitions.



FIGURE 1 - US AIR FORCE MQ-1 PREDATOR DRONE

The US predator drone (above) also features a large relative wingspan of 55 ft (27 ft length) which allows the drone to linger at a cruise altitude of 25 000 ft for 24 hours. The search and rescue aircraft would trade the high cruise altitude for a low altitude to ensure effectiveness in locating people as well as optimizing for range over endurance to maximize the searchable area of the aircraft for each flight [3].

2.2 Initial Design Requirements

After outlining conceptually, the aircrafts main functions and how the aircraft would be constructed, the respective requirements were determined. These requirements are:

TABLE 1 - INITIAL DESIGN REQUIREMENTS [4]

Exterior Dimensions	Value	Units
Height	7.33	ft
Wingspan	30	ft
Length	23.33	ft
Internal Cabin Volume	190	cu. ft
Operating Weights		
Max T/O Weight	2400	lbs
Empty Weight	1225	lbs
Range		
Max Range	1000	Nm
Cruise Altitude	3000	ft
Specific Fuel Consumption	0.28	lbs/hphr
Distances		
Takeoff Distance	775	ft
Landing Distance	1700	ft
Performance		
Rate of Climb	12	ft/s
Max Speed	130	kts
Normal Cruise Speed	123	kts
Stall Speed	50	kts
Payload		
Sensors/Cameras	114.25	lbs
Survival Kit	220	lbs

2.2.1 Weight

The weight of the aircraft is defined at various stages during a flight. The initial or takeoff weight of the aircraft is the weight the fuselage, payload, avionics, powerplant, and fuel. As a starting point, the design assumed the same fuselage and powerplant weight as the Piper PA-28-180 plane. This would serve as a good approximation as the PA-28-180 is a common single-engine aircraft which offers adequate performance and is a simple, proven design.



FIGURE 2 - PIPER PA-28-180 AIRPLANE

The initial weight includes a standard avionics package of control monitoring, communication, navigation equipment etc. The payload assumes a suite of sensors including a Canon EOS 5D DSLR camera, FLIR Boson 640 infrared camera, MicaSense RedEdge-MX multispectral cameras, LiDAR sensors, Synthetic Aperture Radar (SAR), and Uni-pac II survival kit (Qty. 2).

Finally, the initial weight estimate assumes a set amount of fuel based on the aircraft's range requirement which will be discussed in greater detail in following sections.

2.2.2 Internal Payload

The internal payload requirement is a spatial requirement which describes the available internal volumetric space which contains the avionics, sensors, payload, powerplant, and fuel. The initial estimate for this requirement is also based on the internal volume of the PA-28-180.

2.2.3 Cruise speed

The cruise speed of the aircraft is set at a reasonable velocity of 123 knots. This speed is consistent with the cruise speed of the PA-28-180 and allows the aircraft to search a wide area in a short amount of time.

2.2.4 Stall speed

The stall speed of the aircraft is a function of the surrounding air density, the maximum lift coefficient, and the wing-loading. A lower stall speed is optimal to increase the ruggedness of the aircraft in adverse conditions (ex. to avoid stall given strong headwinds due to inclement weather). The initial approximation of 50 kts is based on the PA-28-180.

2.2.5 Range

The range of the aircraft is a critical requirement for the aircraft. A range of 1000 nautical miles was chosen as the target value to provide an effective search radius for the aircraft while minimizing the fuel requirements. This provides a good balance between the functional performance and the adaptability of the aircraft.

Assuming a horizontal field of view (figure 3) of the sensors of approximately 50 degrees and a cruise altitude of 3000 ft, the aircraft would be able to search a total area of 1576 sq. km [Equation 1 and 2].

$$\begin{aligned} \tan \beta &= \frac{l}{h} \\ \tan \left(\frac{50}{2} \right) &= \frac{l/2}{3000 \text{ ft}} \\ l &= 2797.8 \text{ ft} \end{aligned} \quad (1)$$

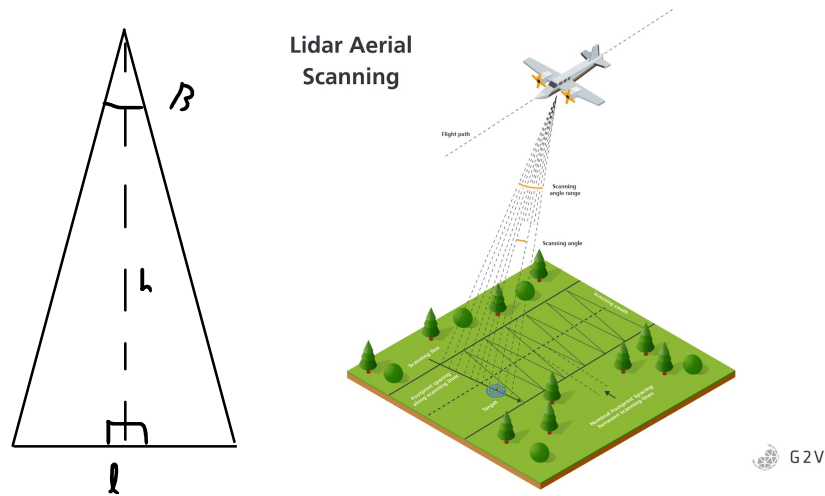


FIGURE 3 - HFOV OF AIRCRAFT SENSORS

$$\begin{aligned} \text{Search Area} &= \text{Range} \cdot l \\ \text{Search Area} &= (2797.8 \text{ ft})(1000 \text{ Nm}) \\ \text{Search Area} &= 1576.9 \text{ km}^2 \end{aligned} \quad (2)$$

2.2.6 Maximum Climb Rates

The maximum climb rate requirement of the aircraft is an estimate based on the performance of the PA-28-180. This value is a function of the excess power provided by the powerplant over the minimum power required during straight, level flight (cruise condition). A lower value is acceptable here as fast climbing is not part of the search and rescue mission set.

2.2.7 Cruise Altitude

The cruise altitude for the aircraft is set at a relatively low 3000 ft due to the specialized application. Search and rescue operations benefit from a lower altitude by allowing a higher resolution of the sensor suite. Additionally, by flying at a lower altitude the people being rescued can potentially signal the aircraft.

This lower altitude has the negative effect of increasing turbulence due to ground effects and increasing the surrounding air density which decreases the overall performance characteristics. These trade-offs are deemed acceptable to accomplish the aircraft's main design objectives.

2.2.8 Takeoff and Landing Distances

Takeoff and landing distance requirements also follow the characteristics of the PA-28-180. A takeoff distance of 775 ft and landing distance of 1700 ft give the aircraft increased flexibility in its effective operating capabilities. The short takeoff distance allows the aircraft to operate out of the smallest regional airports as well as many fields and highways in emergency situations.

2.3 Initial Estimates

2.3.1 Requirement R1 (Initial Weight)

Requirement R1 is the first step in calculating the critical parameters of the aircraft. The goal is to determine the initial weight of the aircraft based on the chosen range and payload as discussed previously. The initial weight (W_o) is broken down by the following equation [Equation 3].

$$\begin{aligned}
 W_o &= W_{crew} + W_{payload} + W_{fuel} + W_{empty} \\
 W_o - W_{fuel} - W_{empty} &= W_{crew} + W_{payload} \\
 W_o \left(1 - \frac{W_{fuel}}{W_o} - \frac{W_{empty}}{W_o} \right) &= W_{crew} + W_{payload} \\
 W_o &= \frac{W_{crew} + W_{payload}}{\left(1 - \frac{W_{fuel}}{W_o} - \frac{W_{empty}}{W_o} \right)}
 \end{aligned} \tag{3}$$

The next step is to break down the weight of the aircraft throughout its flight profile. The respective weights of the aircraft at different points along the profile can be seen in the figure below.

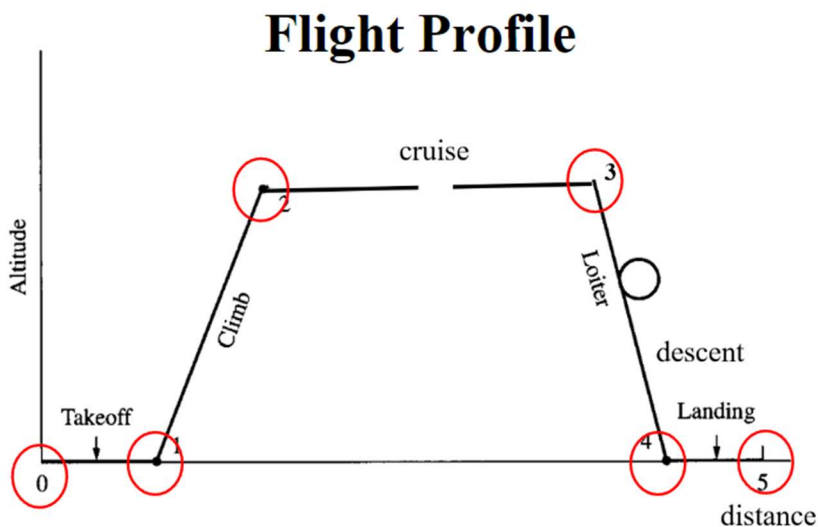


FIGURE 4 - FLIGHT PROFILE OF AIRCRAFT

The weight of the fuel can thus be expressed with the following equation.

$$\begin{aligned}
W_{fuel} &= W_o - W_5 \\
\frac{1}{W_o} (W_{fuel} &= W_o - W_5) \\
\frac{W_{fuel}}{W_o} &= (1 - \frac{W_5}{W_o})
\end{aligned} \tag{4}$$

In which, the ratio of the final weight and the initial weight is:

$$\frac{W_5}{W_o} = (\frac{W_1}{W_o})(\frac{W_2}{W_1})(\frac{W_3}{W_2})(\frac{W_4}{W_5})(\frac{W_5}{W_4}) \tag{5}$$

Now, using the range equation we can solve for the cruise weight ratio assuming a propulsion efficiency of 0.85, a L/D ratio of 14, and the listed fuel consumption and range [5].

$$\begin{aligned}
R &= \frac{n_{pt}}{c} \left(\frac{L}{D} \right) \ln \left(\frac{W_2}{W_3} \right) \\
\frac{W_3}{W_2} &= e^{-\left(\frac{n_{pt}}{c} \left(\frac{L}{D} \right)_{max} \right)} \\
\frac{W_3}{W_2} &= 0.931
\end{aligned} \tag{6}$$

Substituting this value back into equation 4, while assuming the characteristic ratios listed below, we get a W5/W0 ratio of 0.885.

TABLE 2 – FLIGHT PROFILE RATIOS FROM EMPIRICAL DATA [5]

Flight Profile Ratio	Value
$\frac{W_1}{W_o}$	~0.97
$\frac{W_2}{W_1}$	~0.985
$\frac{W_4}{W_5}$	~1
$\frac{W_5}{W_4}$	~0.995
$\frac{W_e}{W_o}$	~0.62

Substituting this value into equation 3 results in a fuel to empty weight ratio of 0.115. This value can be substituted back into the equation 2 to solve for the initial weight of the aircraft:

$$W_o = \frac{W_{crew} + W_{payload}}{\left(1 - \frac{W_{fuel}}{W_o} - \frac{W_{empty}}{W_o} \right)} \tag{3}$$

$$W_o = \frac{334.25 \text{ lbs}}{(1 - 0.115 - 0.62)}$$

$$W_o = 1260.2 \text{ lbs}$$

2.3.2 Requirement R2 (Wing Loading)

The next requirement R2 is based on the critical parameter of wing loading. To find the wing loading we start with an inequality based on the stall speed requirement.

$$V_{stall} = \sqrt{\frac{2}{\rho} \frac{1}{C_{Lmax}} \left(\frac{W}{S} \right)} \leq 50 \text{ kts} \quad (7)$$

Now, substituting values for the surrounding air density and maximum coefficient of lift (empirical data), we solve for the wing loading:

$$\sqrt{\frac{2}{0.002377} \frac{1}{2.34} \left(\frac{W}{S} \right)} \leq 50 \text{ kts} \quad (7)$$

$$18.96 \sqrt{\left(\frac{W}{S} \right)} \leq 84.39 \frac{ft}{s}$$

$$\frac{W}{S} \leq 19.81 \frac{lbs}{ft^2}$$

The next step is to use the equation for landing distance to ensure we are below our requirement.

$$Landing \text{ Distance} = S_a + S_f + S_{fr} + S_g \leq 1700 \text{ ft} \quad (8)$$

Where:

$$\text{i) } S_a = \frac{50 - h_f}{\tan \theta_a} \quad (9)$$

$$\text{ii) } S_f = R \sin \theta_a$$

$$\text{iii) } S_{fr} = 3V_{TD}$$

$$S_{fr} = 3(1.15V_{stall})$$

$$S_{fr} = 3 \left(1.15 \left[18.96 \sqrt{\frac{W}{S}} \right] \right)$$

$$S_{fr} = 65.4 \sqrt{\frac{W}{S}}$$

$$\text{iv)} \quad S_g = \frac{(1.15)^2 \left(\frac{W}{S}\right)}{g \cdot \rho \cdot C_{Lmax} \cdot \mu_r}$$

v)

Now solving for the radius R, assuming n = 1.2:

$$\begin{aligned} R &= \frac{(V_f)^2}{g(n-1)} \\ R &= \frac{(1.23 \cdot V_{stall})^2}{g(n-1)} \\ R &= \frac{(1.23 \cdot 84.39)^2}{32.2(1.2-1)} = 1673.1 \text{ ft} \end{aligned} \quad (10)$$

Solving for the flare height, assuming the angle of approach is 3 degrees:

$$\begin{aligned} h_f &= R(1 - \cos\theta_a) \\ h_f &= 1673.1(1 - \cos[3^\circ]) = 2.3 \text{ ft} \end{aligned} \quad (11)$$

Finally, substituting all of these values back into equation 7:

$$\begin{aligned} \text{Landing Distance} &= S_a + S_f + S_{fr} + S_g \leq 1700 \text{ ft} \\ \frac{50 - h_f}{\tan\theta_a} + R\sin\theta_a + 65.4 \sqrt{\frac{W}{S}} + \frac{(1.15)^2 \left(\frac{W}{S}\right)}{g \cdot \rho \cdot C_{Lmax} \cdot \mu_r} &\leq 1700 \text{ ft} \\ 910.3 + 87.6 + 291.1 + 365.6 &\leq 1700 \text{ ft} \\ 1654.6 &\leq 1700 \text{ ft} \end{aligned} \quad (7)$$

2.3.3 Requirement R3 (Thrust-to-Weight Ratio)

The purpose of section R3 is to determine the power required in three different key areas of flight. Then use the greatest of the three values to help select a power plant. The three flight conditions are, takeoff distance, maximum rate of climb and maximum velocity.

First, we will find the power required for a minimum takeoff distance of 775ft to clear a 50ft obstacle.

$$\text{Takeoff Distance} = S_g + S_a \leq 775 \quad (12)$$

$$(Cl)_{max} = 0.9(2.2)$$

$$(Cl)_{max} = 1.98$$

$$Sg = \frac{1.21 \cdot \left(\frac{w}{s}\right)}{32.17 \cdot 0.002377 \cdot 1.98 \cdot \left(\frac{T}{w}\right)}$$

$$Sg = \frac{158.23}{\left(\frac{T}{w}\right)}$$

$$R = \frac{6.96 \cdot V_{stall}^2}{g}$$

$$R = \frac{6.96 \cdot 84.39^2}{32.17}$$

$$R = 1540.78$$

$$\theta_{ob} = \cos^{-1}\left(1 - \frac{50}{1540.78}\right)$$

$$\theta_{ob} = 14.636$$

$$Sa = R \cdot \sin(\theta_{ob})$$

$$Sa = 1540.78 \cdot \sin(14.636)$$

$$Sa = 389.32$$

$$\left(\frac{T}{w}\right) = 0.4102$$

$$V_{\infty(takeoff)} = 0.7 \cdot 1.1 \cdot V_{stall}$$

$$V_{\infty(takeoff)} = 0.7 \cdot 1.1 \cdot 84.39$$

$$V_{\infty(takeoff)} = 64.98$$

$$P = \frac{T \cdot V_{\infty(takeoff)}}{n_{pr}}$$

$$P = \frac{0.4102 \cdot W_o \cdot V_{\infty(takeoff)}}{n_{pr}}$$

$$P = \frac{0.4102 \cdot 1260 \cdot 64.98}{0.85}$$

$$P_{hp} = \frac{0.4102 \cdot 1260 \cdot 64.98}{0.85 \cdot (550)}$$

$$P_{hp} = 71.84$$

Here are the calculations to determine the minimum power required to achieve the max rate of climb we set as a parameter.

$$\begin{aligned}
C_{fe} &= 0.0043 \\
C_{D,0} &= \frac{S_{wet}}{S} \cdot C_{fe} \\
C_{D,0} &= \frac{S_{wet}}{S} \cdot 0.0043 \\
C_{D,0} &= 0.017
\end{aligned} \tag{13}$$

$$C_D = C_{D,0} + k \cdot C_t^2$$

$$\begin{aligned}
K &= \frac{1}{4 \cdot C_{D,0} \cdot \left(\frac{L}{D}\right)_{max}^2} \\
K &= \frac{1}{4 \cdot 0.017 \cdot (14.2)^2} \\
K &= 0.0729
\end{aligned}$$

$$e_o = 0.6$$

$$\begin{aligned}
AR &= \frac{1}{\Pi \cdot e_o \cdot K} \\
AR &= 7.227
\end{aligned}$$

$$\begin{aligned}
\frac{n_{pt} \cdot P}{W} &= \left(\frac{R}{C}\right)_{max} + \left(\frac{2}{\rho} \cdot \left(\frac{K}{3 \cdot C_{D,0}}\right)^{.5} \cdot \frac{W}{s}\right)^{.5} \cdot \frac{1.155}{\left(\frac{L}{D}\right)_{max}} \\
\frac{n_{pt} \cdot P}{W} &= \left(12 + \left(\frac{2}{0.002377} \cdot \left(\frac{0.0729}{3 \cdot 0.017}\right)^{.5} \cdot 19.806\right)^{.5} \cdot \frac{1.155}{14.2}\right) \\
\frac{n_{pt} \cdot P}{W} &= 23.48
\end{aligned}$$

$$\begin{aligned}
P_{hp} &= \frac{23.48 \cdot (1260.158)}{0.85} \\
P_{hp} &= 63.29
\end{aligned}$$

Here are the calculations to find the minimum horsepower required to achieve the maximum velocity we set as a parameter.

$$\begin{aligned}
T = D &= \left(\frac{1}{2} \cdot \rho_{\infty} \cdot V_{\infty}^2 \cdot S \cdot C_{D,0}\right) + \left(\frac{2 \cdot K \cdot S}{\rho_{\infty} \cdot V_{\infty}^2} \cdot \left(\frac{W}{s}\right)^2\right) \\
\frac{T}{W} &= \left(\frac{1}{2} \cdot \rho_{\infty} \cdot V_{\infty}^2 \cdot \frac{C_{D,0}}{\frac{W}{s}}\right) + \left(\frac{2 \cdot K}{\rho_{\infty} \cdot V_{\infty}^2} \cdot \frac{W}{s}\right)
\end{aligned} \tag{14}$$

$$\frac{w_2}{w_o} = \frac{w_1}{w_o} \cdot \frac{w_2}{w_1}$$

$$\frac{w_2}{w_o} = 0.97 \cdot 0.985$$

$$w_2 = 0.955 \cdot w_o$$

$$w_2 = 0.955 \cdot 1260.158$$

$$w_2 = 1204.02$$

$$w_{mc} = w_2 - \left(\frac{w_2 - w_3}{2} \right)$$

$$\frac{w_3}{w_2} = \frac{1}{1.119} = 0.893$$

$$\frac{w_{mc}}{w_2} = \frac{1 + .893}{2} = 0.9465$$

$$w_{mc} = 1139.605$$

$$\frac{w_{mc}}{s} = \frac{1139.605}{60.89} = 18.717$$

$$\frac{T}{W_{mc}} = 0.07429$$

$$p = \frac{1}{n_{pt}} \cdot \frac{T}{W_{mc}} \cdot w_{mc} \cdot V_{max}$$

$$p_{hp} = \frac{\left(\frac{1}{0.85} \cdot .07429 \cdot 1139.605 \cdot 219.4 \right)}{550}$$

$$p_{hp} = 39.73$$

3.0 Design Procedure

This project followed the design process outlined during the course. The first step was to determine the first estimate of the aircraft's takeoff weight. This was done with empirical data as well as specifications derived from the piper PA-28-180, which shared a similar design to our target aircraft. This weight was then used to calculate specific parameters such as wingspan and power requirements in the first iteration so that an engine and both wing and tail details could be determined. At the end of the first iteration, a more accurate takeoff weight was estimated based on the derived details of the aircraft. This weight was then used as the new initial estimate, and another iteration was completed as outlined above. This process was repeated until the calculated take-off weight converged with the initial estimate, which took a total of 6 iterations.

FIGURE 5: DESIGN PROCEDURE PROCESS FLOW

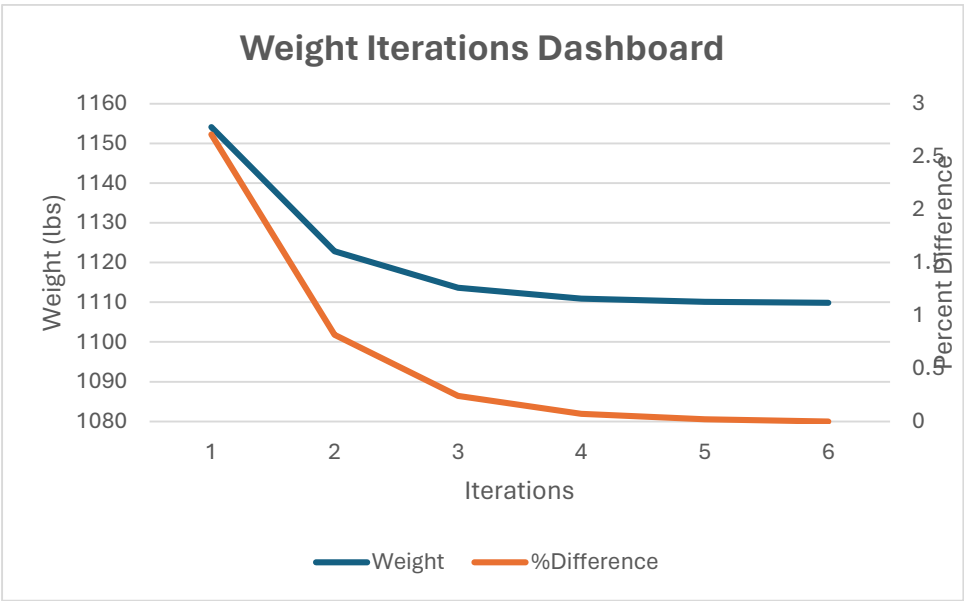
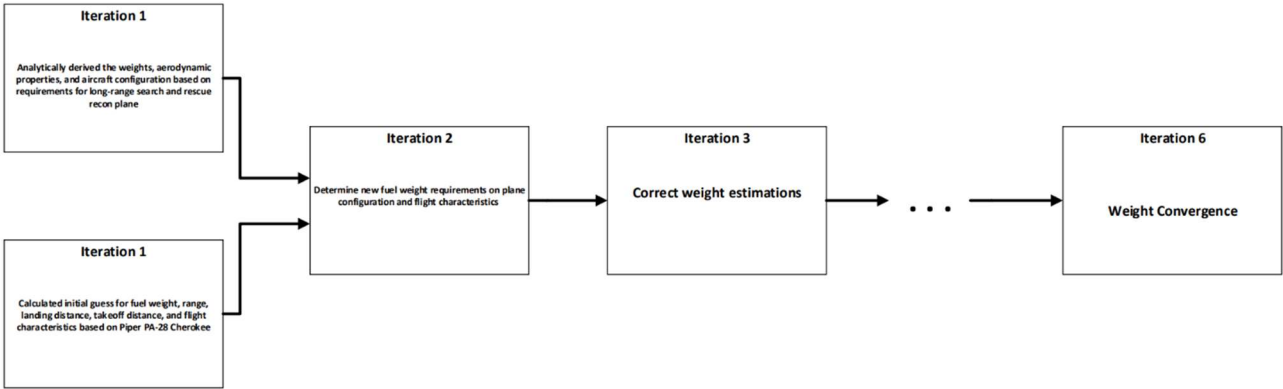


FIGURE 6: WEIGHT ITERATIONS DASHBOARD

4.0 Configuration

The configuration of the newly designed aircraft was significantly influenced by the Piper PA-28 Cherokee, known for its reliability and adaptability. However, substantial deviations were incorporated to enhance performance specifically for low-altitude operations essential in search and rescue and offshore reconnaissance missions. These modifications aimed to improve maneuverability, stability, and visibility beneath cloud cover, crucial for the specific mission profiles envisioned.

Adjustments to the fuselage, wing, tail, and powerplant were meticulously designed to optimize the aircraft for low-altitude flight dynamics, where environmental factors and air density present unique challenges. The center of gravity and aerodynamic center were strategically modified to enhance stability and control. The fuselage was reshaped to improve aerodynamics and operational efficiency, while the wings and tail were redesigned for better low-speed performance and maneuverability, ensuring the aircraft remains robust and responsive during diverse operational conditions.

4.1 Fuselage Design

4.1.1 Internal Fuselage Subsystems

The fuselage design of the search and rescue aircraft, while drawing inspiration from the Piper PA-28 Cherokee, is tailored specifically with the mission's requirements in mind. Like the Cherokee, the fuselage adopts a narrow tubular structure, primarily to accommodate avionics, a payload bay, the single engine, and the central fuel transfer system. Table 3 detailed presents the components involved in each system, and Figure 7 illustrates the outer mold lines of the fuselage along with the strategic placement of these systems.

TABLE 3: SUBSYSTEMS COMPONENTS

Avionics	
System	Weight [lbs]
Canon EOS 5D Mark IV DSLR Camera	2
Infrared (IR) Camera FLIR Boson 640	0.15
Multispectral Cameras MicaSense RedEdge-MX	0.8
LiDAR	1.3
Synthetic Aperture Radar (SAR) ceye X3 (Miniature SAR Satellite)	110
Payload	
System	Weight [lbs]
uni-pac ii (x2)	220
Powerplant	
System	Weight [lbs]
Viking 90	220.6

Fuel Transfer System	
System	Weight [lbs]
Cessna 172 Fuel Transfer System	15.2

Positioning the avionics at the front of the fuselage was a deliberate choice aimed at maximizing visibility for the aircraft's multiple camera systems and sensors essential for reconnaissance tasks. This fixed load at the front is strategically balanced by the rear-placed engine, which helps distribute the weight across the aircraft. The dynamic changing loads, including the payload and the central fuel transfer system, are situated near the center of the aircraft, close to the center of gravity. This placement is critical as it minimizes the impact on stability when these weights shift during various phases of a mission. By thoughtfully arranging these components, the design enhances the aircraft's operational efficiency and safety, ensuring that it remains stable and responsive under varying load conditions.

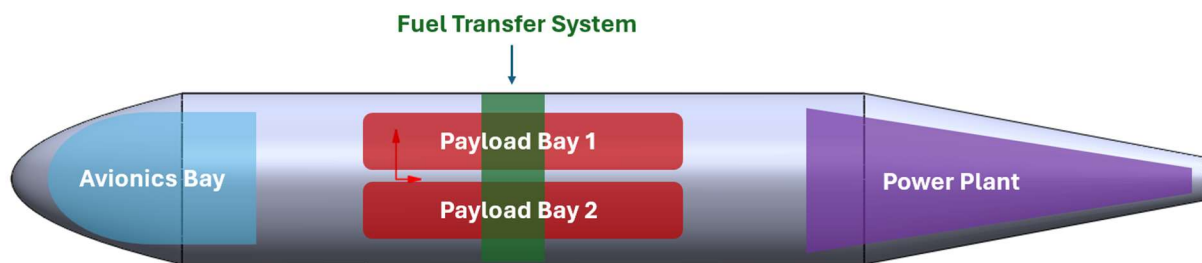


FIGURE 7: FUSELAGE LAYOUT

4.1.2 Master Outer Module Lines

The design of the fuselage not only supports the operational components and load distribution but also significantly contributes to the lift characteristics of the aircraft. This contribution is influenced by the wetted area of the fuselage, which plays a vital role in the aerodynamic performance. The fuselage is constructed from three main cross-sectional areas: the elliptical nose cone, the tubular main fuselage, and the trapezoidal-cone tail. These shapes were chosen not only for their structural benefits and capacity to house necessary systems but also for their aerodynamic properties.

The elliptical nose cone is designed to minimize aerodynamic resistance and improve air flow around the fuselage, while the tubular main section provides an optimal balance between internal volume and surface area, reducing drag and enhancing lift. The trapezoidal-cone tail section assists in smoothly tapering off the airflow, further minimizing drag and potentially improving the aircraft's stability and control. The calculation of the wetted area, which can be seen below, takes into account these varied shapes and their respective contributions to the aircraft's overall aerodynamic efficiency. This thoughtful integration of form and function ensures that the fuselage design not only meets the spatial and operational requirements but also enhances the lift and flight characteristics essential for effective search and rescue operations.

$$S_{Wetted} = \sum (S_{Elliptical\ Dome} + S_{Circular\ Cylinder} + S_{Trapazoidal-Cone}) \quad (15)$$

$$S_{Elliptical\ Dome} = \frac{\pi ab}{4} = \frac{\pi \cdot 3 \cdot 3}{4} \quad (16)$$

$$S_{Elliptical\ Dome} = 7.07\ ft^2$$

$$S_{Circular\ Cylinder} = lw = 12 \cdot 3 \quad (17)$$

$$S_{Circular\ Cylinder} = 36\ ft^2$$

$$S_{Trapazoidal-Cone} = \left(\frac{a+b}{2}\right)l = \left(\frac{3+0.75}{2}\right) \cdot 6 \quad (18)$$

$$S_{Trapazoidal-Cone} = 11.25\ ft^2$$

$$S_{Wetted} = 54.32\ ft^2$$

4.2 Powerplant

For the powerplant configuration of the search and rescue aircraft, a pusher design was selected. This decision was driven by the need to accommodate extensive avionics systems, including cameras, LIDAR, and sensors crucial for locating debris, boats, life rafts, and individuals during missions. Positioning the single engine at the rear of the aircraft allows the front section to be utilized for a clear housing dedicated to these systems. This arrangement not only maximizes the use of space but also avoids the aerodynamic drag penalties associated with external bulges and housings that would otherwise be necessary for these components.

The pusher configuration offers several significant advantages. Firstly, it ensures higher-quality airflow over the wing and fuselage, crucial for the efficient operation of onboard sensors. Additionally, the inflow to the rear-mounted propeller induces a favorable pressure gradient at the fuselage's rear, allowing for a steeper fuselage closure without flow separation. This design enables a shorter fuselage, reducing the wetted surface area and potentially lowering drag. Engine noise in the cabin area is also diminished, and the pilot benefits from an unobstructed frontal field of view, enhancing operational safety and effectiveness. However, this configuration does shift the center of gravity rearward, potentially reducing longitudinal stability, and poses risks of propeller damage from flying debris during landing and challenges in engine cooling. These disadvantages were carefully evaluated against the operational gains, leading to the conclusion that the pusher configuration's benefits align well with the mission requirements of the aircraft [5].

After the first iteration of the design process, it was determined that the minimum power required for the aircraft was 86.7 HP. Since, the power requirements of the aircraft were predicted to decrease with future design iterations, a factor of safety for available power wasn't considered, and a 90 HP Viking Aircraft engine was selected. This engine is built off the Mitsubishi Mirage 1.2 litre, 3-cylinder engine architecture and features fuel injection and a gearbox reduction. This engine was selected due to its low cost, light weight and small form factor. While Viking does offer turbocharged engines, a non-turbocharged model was selected due to the low altitude this aircraft

would be operating at. The dry weight of the selected engine is 159 lbs, and the peak power is 90 HP at 5800 rpm [6].

Since this engine is predominantly used in the experimental aircraft industry, little data is provided by the manufacturer regarding specific fuel consumption. However, it is safe to say that this engine would be using less fuel than that of the Piper PA-28-180 which the initial design estimates were based on. For this reason, the fuel consumption of the Piper PA-28-180 was used throughout the design process, essentially overestimating the fuel consumption of this smaller engine. This approximation would lead to an increase in the actual range of the aircraft.

4.3 Aerodynamics

4.3.1 Aircraft Wings

4.3.1.1 Wing Design

The wing design of the search and rescue aircraft is modeled after the Piper PA-28 Cherokee, utilizing the NACA 65(2)-415 airfoil. This particular airfoil is a member of the NACA 6-series, renowned for its ability to precisely control the location of minimum pressure points and effectively achieve optimal lift coefficients. These airfoils are strategically designed with a focus on critical performance metrics such as lift and drag, making them highly suitable for high-speed applications. The NACA 65(2)-415 airfoil features a maximum thickness of 15% at 39.9% of the chord and a maximum camber of 3% at 45% of the chord [7]. The thicker max camber of this airfoil allows for increased fuel storage, a crucial factor for long-range search and rescue operations.

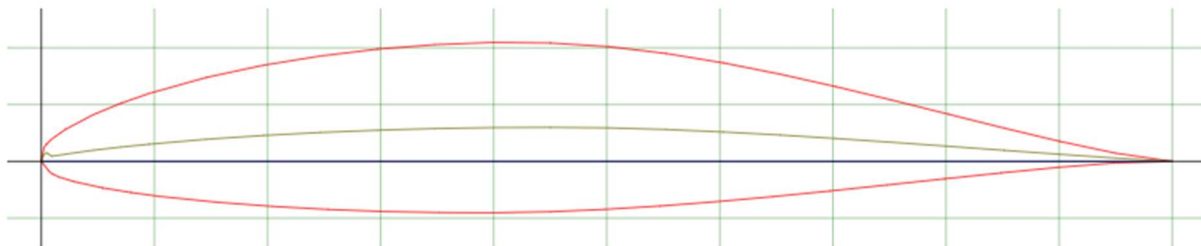


FIGURE 8: NACA 65(2)-415 AIRFOIL

The decision to implement a wing with a 50% taper ratio was influenced by several aerodynamic considerations. A tapered wing typically reduces induced drag more effectively than wings with a constant chord. This reduction is attributed to the wing's closer approximation to an optimal elliptical lift distribution, which minimizes downwash and the associated drag at the wingtips. Moreover, a tapered wing allows for operation at lower lift coefficients, enhancing flight efficiency. This wing design also affects landing behavior, facilitating smoother approaches and landings due to its modified aerodynamics.

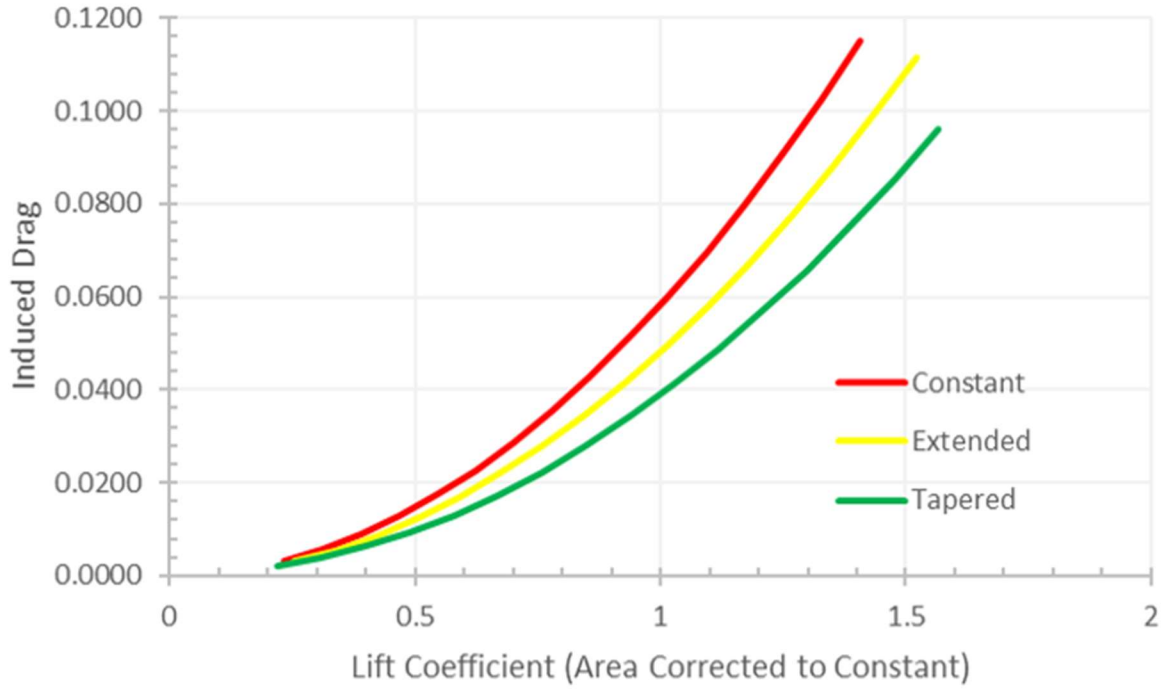


FIGURE 9: INDUCED DRAG ESTIMATE FOR PA-28 SERIES USES THE NACA 65(2)-415 AIRFOIL

The aspect ratio was determined based on the coefficient of drag equation and the airplane efficiency coefficient, anticipated to be similar to the Cherokee, at $e=0.6$. This parameter allowed for the reworking of the equation to establish an appropriate aspect ratio and the critical geometric properties of the wings such as root chord length, tip chord length, wing area, and mean aerodynamic chord.

$$AR = \frac{1}{\pi \cdot e \cdot k} = 7.27 \quad (19)$$

$$S = \left(\frac{W_0}{\left(\frac{W}{S} \right)} \right) \quad (20)$$

$$b = \sqrt{(AR \cdot S)} \quad (21)$$

$$\lambda = 0.5 \quad (22)$$

$$c_r = \frac{2S}{(1 + \lambda)b} \quad (23)$$

$$c_t = \lambda c_r \quad (24)$$

$$\bar{c} = \frac{2}{3} c_r \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right) \quad (25)$$

4.3.1.2 Low Wing Configuration

In the design of the search and rescue aircraft, a low-wing configuration was selected primarily due to its advantages in the integration and functionality of the landing gear. This configuration permits the landing gear to be retracted directly into the wing box, which is among the strongest structural elements of the aircraft. This arrangement simplifies the landing gear mechanics and enhances the aircraft's structural integrity during the critical phases of landing and gear retraction. Additionally, this design provides the necessary ground clearance for the engine or propeller, ensuring that the landing gear maintains proper height above the ground to prevent potential operational interferences.

Furthermore, the low-wing configuration necessitates the implementation of dihedral to ensure lateral stability and incorporates a fillet at the wing-body juncture to mitigate aerodynamic interference. This fillet effectively minimizes adverse pressure gradients and consequent flow separation, which can lead to increased drag and unstable buffeting. Consequently, these design considerations enhance the aircraft's aerodynamic efficiency and stability, making the low-wing configuration particularly suitable for meeting the operational demands and safety requirements of search and rescue missions.

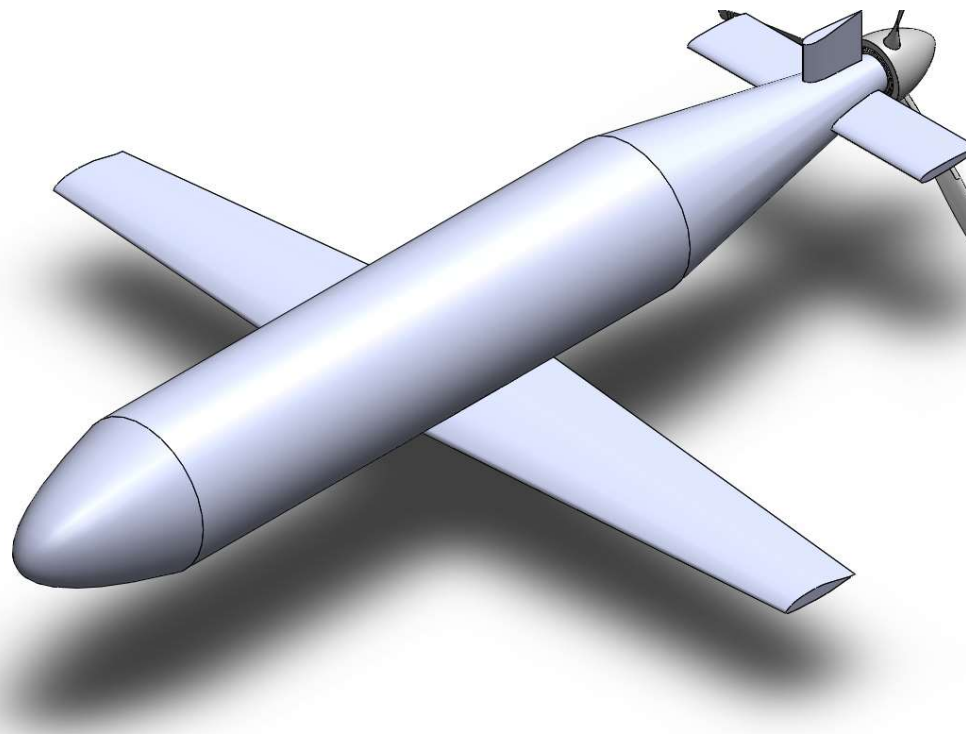


FIGURE 10: LOW-WING CONFIGURATION

4.3.1.3 High-Lift Devices

Considering the requirements for enhanced lift during lower-speed operations typical in search and rescue missions, the wing design incorporates double-slotted flaps. This choice of high-lift device offers significant advantages, particularly during the critical phases of takeoff and landing. Double-slotted flaps are highly effective in increasing the wing surface area and camber, which boosts the lift coefficient substantially without a corresponding increase in drag. This enables the

aircraft to maintain slower approach speeds and shorter landing distances, essential for operations in diverse and potentially constrained environments.

These flaps work by directing the airflow over the wing, maintaining smoother airflow at higher angles of attack, thus delaying stall and increasing safety margins during low-speed maneuvers. The use of double-slotted flaps, therefore, not only enhances the aircraft's aerodynamic efficiency but also improves operational flexibility and safety during the unique and challenging conditions encountered during search and rescue missions [5].

For the NACA 65(2)-415 airfoil selected to be used in the design, empirical data exists for the increase in the lift coefficient caused by deploying double slotted flaps a deflection angle of 55 degrees. This can be seen in the schematics and charts shown in the following three figures.

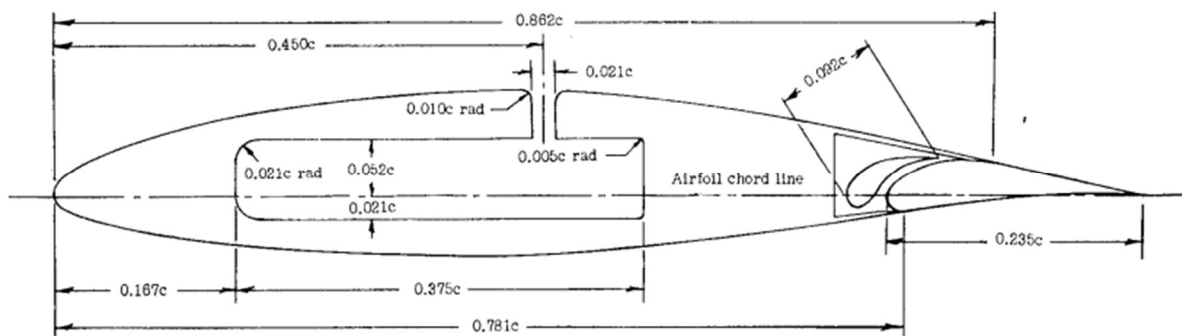


FIGURE 11: AIRFOIL WITH FLAP RETRACTED

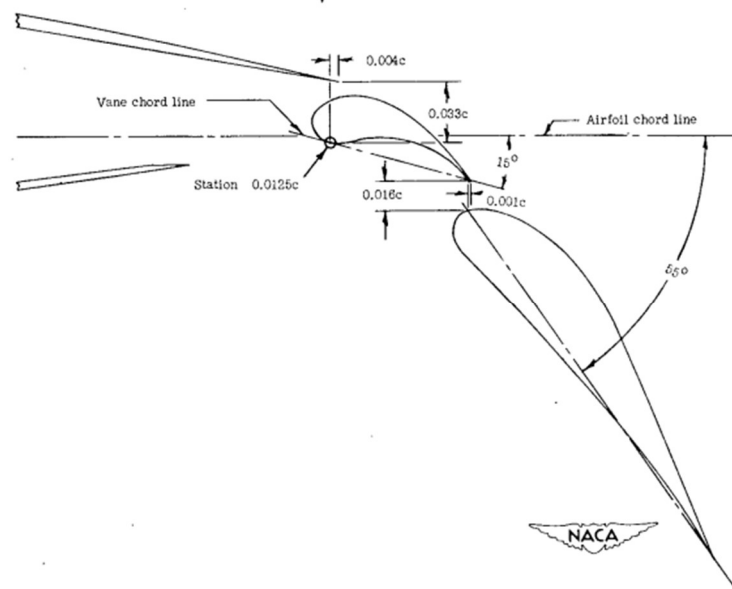
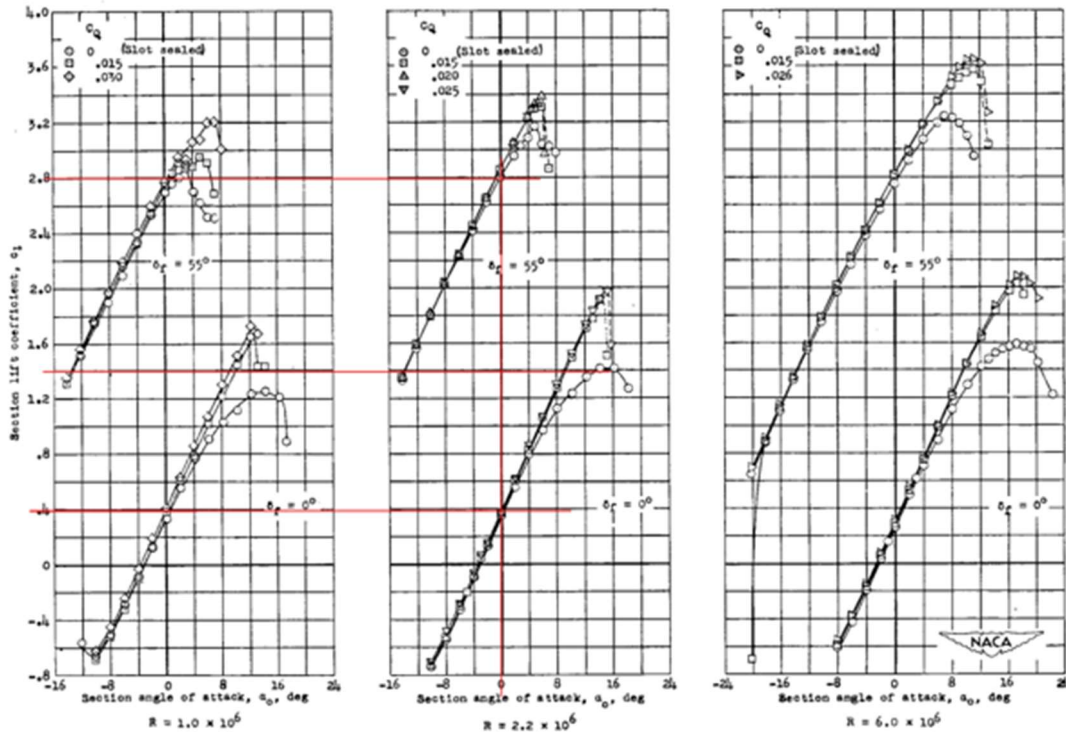


FIGURE 12: DETAILS OF FLAP FOR A DEFLECTION OF 55°



(a) Model in smooth condition.

Figure - Section lift characteristics of the NACA 65₂-415 airfoil section with a double slotted flap and a boundary-layer control slot at 0.45c.

FIGURE 13: INCREASE IF LIFT COEFFICIENT ON NACA 65(2)-415 AIRFOIL DUE TO DOUBLE SLOTTED FLAP AT 55DEG [9]

From the data, we found that using a double slotted flap on the NACA 65(2)-415 airfoil resulted in an increase of 2.4 at a deflection angle of 55 degrees. However, for the purpose this aircraft design, smaller deflection angles were chosen to increase the lift during take off and drag. These flaps with smaller deflection angles would be easier to manufacture and reduce the flow disruption around the wing. The deflection angles for take off and landing as well as the interpolated lift coefficient increases can be seen in the following table. The data points were taken from the middle chart for reynolds number of 2.2×10^6 as that was closest to our flow case. The lift coefficients were interpolated using the following formula.

$$\Delta(c_l)_{\max, \delta_f = x} = 2.4 \left(\frac{(\delta_f = x)}{55} \right) \quad (26)$$

TABLE 4: TAKE OFF AND LANDING LIFT COEFFICIENTS

	DELTA_CL	ANGLE		
DATA	2.4	55	Cl,max	1.4
TAKEOFF	0.65	15	Cl,max take off	2.05
LANDING	1.53	35	Cl,max landing	2.93

4.3.2 Center of Gravity

In the design process of the search and rescue aircraft, determining the center of gravity (CG) is a critical step that follows the selection of the fuselage and wing configuration. This calculation is essential for defining the placement of these components and designing the tail structure. The center of gravity calculation facilitates subsequent analyses on stability and allows for the reconfirmation of wing placement, ensuring that the aircraft maintains optimal aerodynamic properties and stability throughout its operational envelope.

4.3.2.1 Aerodynamic Center of Fuselage Section

The weight distribution within the fuselage, encompassing the avionics bay, payload, and engine, was meticulously evaluated to determine the aerodynamic center of the fuselage section. These fixed loads were strategically positioned, and their distances were iteratively adjusted to align the aerodynamic center centrally. This iterative process ensures that the aerodynamic center supports optimal flight characteristics and structural balance. The mathematical representation of this configuration is detailed in the equation provided below:

$$x_{ac}^* = \frac{\sum w_i x_{cg,i}}{\sum w_i} \quad (27)$$

4.3.2.2 Center of Gravity of Aircraft

For stability, the most advantageous configuration positions the aerodynamic center slightly forward of the center of gravity, with the wings placed rearward relative to the aircraft's center of gravity. To accurately place the wings and ensure they contribute positively to the overall stability of the aircraft, it is first necessary to estimate the weight of the wings. This estimation can be calculated using the imperial estimation method, which provides a reliable approximation of wing weight as shown in the following equation:

(28)

$$S_{Exposed} = 2 \sqrt{\frac{w_{wing} \approx 2.5 S_{Exposed}}{\left(\frac{c_{fuselage} - c_t}{2}\right)^2 + \left(\frac{b}{2} - \frac{b_{fuselage}}{2}\right)}} \quad (29)$$

With the weight of the wings determined, their placement becomes a direct function of the center of gravity. This placement is chosen iteratively, based on the center of gravity calculation, to ensure that the aircraft maintains balance and stability under various flight conditions and load configurations. The equation used for determining the center of gravity, taking into account the contributions from all major components including the wings, is outlined below:

$$x_{wings} = x_{ac}^* + 0.15\bar{c} \quad (30)$$

$$x_{cg} = \frac{\sum w_i x_{cg,i}}{\sum w_i}, \text{ wings included} \quad (31)$$

Through these calculations, the design team can accurately configure the aircraft for optimal performance, ensuring that all components are harmoniously integrated to support the mission-specific requirements of search and rescue operations.

4.3.3 Aircraft Tail

4.3.3.1 Tail Configuration

The tail configuration chosen for the search and rescue aircraft is a conventional tail, positioned centrally at the tail end of the fuselage as depicted in Figure 14. Among the variety of tail designs, the conventional tail was selected due to its optimal balance of structural weight, stability, and control capabilities. A conventional tail design ensures that during stall, the wake from the horizontal tail does not obscure the rudder on the vertical tail, which is crucial for maintaining control integrity under various flight conditions. Although other configurations like the T tail provide benefits such as reduced induced drag on the vertical tail and prevention of rudder blanketing at stall, these advantages come with increased structural requirements and weight [5].

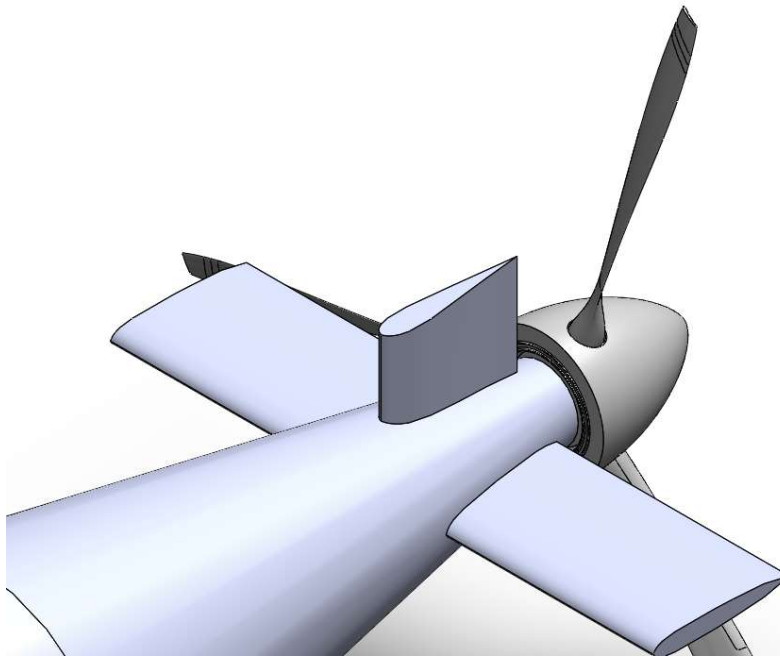


FIGURE 14: CONVENTIONAL TAIL CONFIGURATION

For the specific design of this aircraft, the conventional tail was selected due to its lighter weight structure and its proven reliability in providing reasonable stability and control. The fuselage length of 20 ft assists in determining the placement of the horizontal tail, which is positioned 19.94 ft from

the nose to align with the aerodynamic center, ensuring effective leverage and control surfaces interaction.

4.3.3.2 Horizontal and Vertical Tail Size

The sizing of the tail components, both horizontal and vertical, is among the less precise yet critically important aspects of aircraft design. The primary function of the horizontal tail, equipped with the elevator, is to ensure longitudinal stability, control, and trim. Conversely, the vertical tail, with its rudder, provides directional (yawing) stability and control. The dimensions of these tails must be adequately scaled to offer necessary stability and control capabilities under all operational conditions.

The sizing process for both tail components is largely empirical, based on historical data and not detailed analytical models. This approach aligns with the design philosophy of balancing empirical evidence with practical design needs without delving into overly complex calculations. Tail volumes are defined by ratios, where V_{HT} and V_{VT} represent the horizontal and vertical tail volume ratios, respectively. These ratios are calculated using the distance between the center of gravity of the aircraft and the aerodynamic centers of the tails, alongside the planform and sideview areas of the tails. For this aircraft, the recommended values from historical single-engine general aviation data are $V_{HT}=0.7$ and $V_{VT}=0.04$ (Raymer, Ref. 25), which will be applied in our design to ensure adequate stability and control [5].

$$AR_{HT} = 4 \quad (32)$$

$$AR_{VT} = 1.5 \quad (33)$$

$$S_{HT} = \frac{V_{HT} \bar{c} S}{l_{HT}} \quad (34)$$

$$b_{HT} = \sqrt{(AR S_{HT})} \quad (35)$$

$$S_{VT} = \frac{V_{VT} b S}{l_{VT}} \quad (36)$$

$$h_{VT} = \sqrt{(AR S_{VT})} \quad (37)$$

4.3.4 Longitudinal Stability Analysis

To verify that the search and rescue aircraft maintains stability and is capable of autonomously returning to level flight following a disturbance, a longitudinal stability analysis was performed. This analysis is fundamental to ensuring that the aircraft can operate safely and efficiently, especially under the high workload conditions typical of search and rescue missions.

Longitudinal stability refers to the aircraft's ability to restore itself to its original flight path after an external disturbance affects its angle of attack. The analysis focused on ensuring that the moment coefficient about the center of gravity has a negative slope when plotted against changes in the angle of attack. A negative slope indicates that any increase in angle of attack generates a restoring moment that pushes the aircraft back towards its initial state, thus ensuring stability.

The stability analysis involved calculating the aircraft's pitching moment coefficient in relation to changes in angle of attack. This calculation was crucial in determining whether the aircraft would behave in a stable manner under different flight conditions. The negative relationship required between the moment coefficient and the angle of attack was confirmed through this analysis, dictating the proper placement of wings and control surfaces relative to the fuselage.

$$a = \frac{2\pi}{1 + \frac{2\pi}{\pi A R e}} \quad (38)$$

$$a = \frac{2\pi}{1 + \frac{2\pi}{\pi \cdot (7.27) \cdot (0.6)}}$$

$$a = 4.309$$

$$\frac{dC_{M,CG}}{d\alpha} = a \left[\frac{l_{w-HT} \bar{c}}{\bar{c}} \right] \quad (39)$$

$$\frac{dC_{M,CG}}{d\alpha} = 4.309 \left[\frac{0.7 \cdot 2.57 - 0.7 \cdot 2.57 \cdot 2.57}{2.57} \right]$$

$$\frac{dC_{M,CG}}{d\alpha} = -4.747$$

The results from this analysis were instrumental in finalizing the aircraft configuration, particularly concerning the positioning of the horizontal tail and the adjustment of control surfaces. By validating that the moment coefficient behaved as expected, the design team could ensure that the aircraft would not only meet operational standards but also enhance safety during missions. This thorough examination was necessary to establish that the aircraft possesses inherent stability, crucial for its intended search and rescue operations.

4.3.5 Landing Gear

In the design of the landing gear for the search and rescue aircraft, a tricycle-style configuration was chosen, as illustrated in Figure 15. This configuration was selected due to several significant advantages: it ensures that the cabin floor remains horizontal when the aircraft is on the ground, improving both passenger comfort and cargo management. Additionally, this configuration enhances forward visibility for the pilot while on the ground and improves stability during ground operations, allowing the aircraft to effectively "crab" into crosswinds without needing to align the fuselage parallel to the runway.

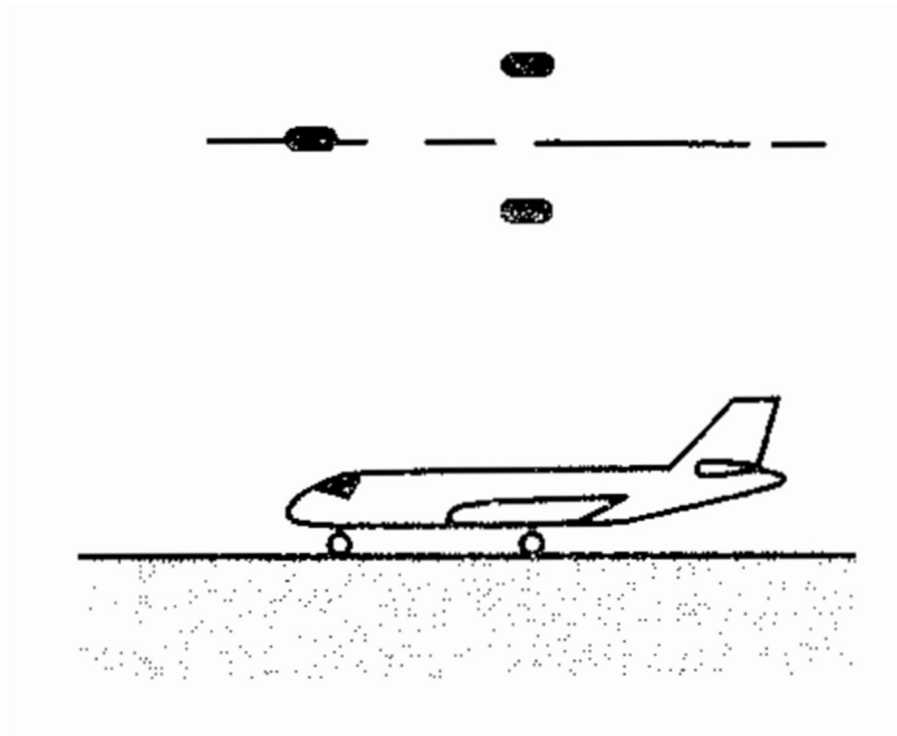


FIGURE 15: TRICYCLE LANDING GEAR

The center of gravity (CG) of the aircraft is strategically positioned ahead of the main wheels, a requirement in the tricycle configuration that further enhances stability during ground rolls. The placement of the landing gear and its integration into the wing structure were carefully considered. Given that the landing gear retracts into the wing, the wing itself needed to be located relative to the fuselage in such a way that accommodates this mechanism while supporting optimal weight distribution and aircraft balance. Initially, the wing was positioned to align the mean aerodynamic center with an early estimate of the CG at 10.20 ft; this was later adjusted to 10.59 ft to refine the balance as the design progressed. This iterative process is vital for confirming the final placement of the wing and ensuring that the aircraft's aerodynamic center lies appropriately behind the CG for enhanced longitudinal stability.

4.3.6 Scale Model of Aircraft

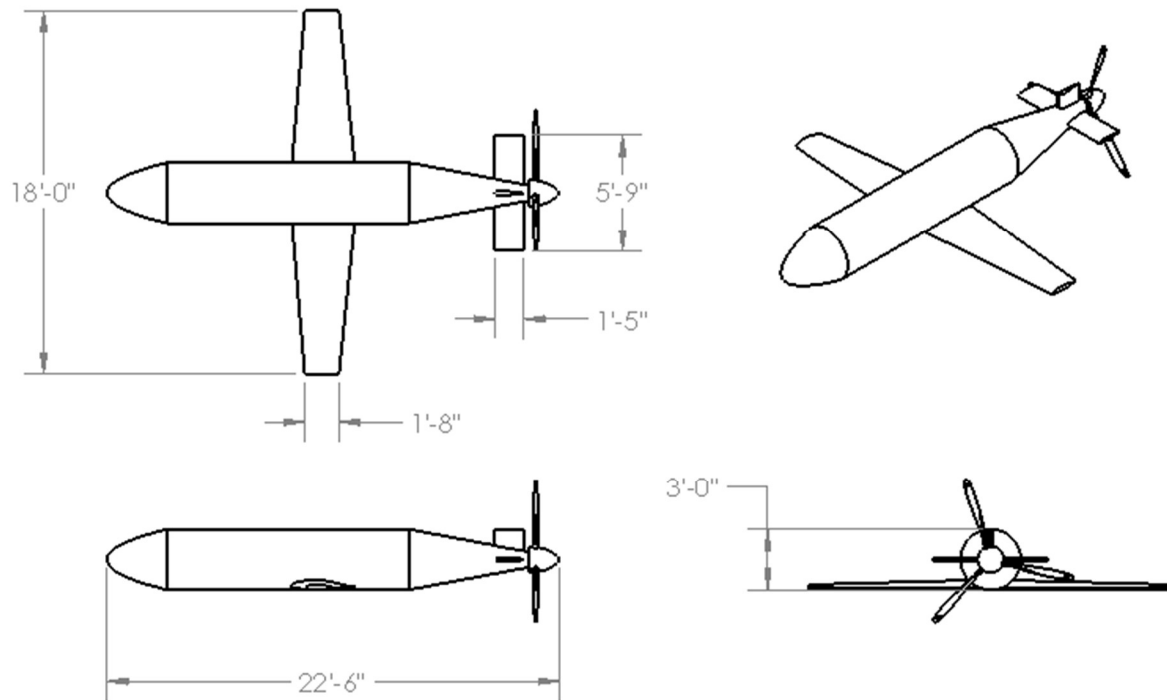


FIGURE 16: OVERALL DIMENSIONS AND STRUCTURE OF AIRCRAFT

5.0 Performance Analysis

Our estimate of weight at the beginning and for the first set of calculations was $W_0=1260.16$ lbs, after the final design iteration the aircraft weight estimate is now $W_0=110.14$. This did not change our wing loading estimate which is equal to $W_0= 24.78$ lb/ft².

The aerodynamic coefficients did not change across the iterations intentionally, however in a more comprehensive aircraft design iteration this could be taken into account.

$$C_{D,o} = 0.017$$

$$C_{Lmax} = 2.93$$

$$\left(\frac{L}{D}\right)_{max} = 14.2$$

Below in Table 5 is a table showing the minimum power required in horsepower for each flight condition across the first calculation and after the last iteration. As you can see below the power required for each flying condition decreased our iterations, which makes sense considering the total weight decreased while the aerodynamic coefficients remained constant.

TABLE 5: CHANGE IN MINIMUM POWER REQUIRED ACROSS FIRST AND LAST ESTIMATION

	First estimate	Final estimate
Take off distance requirement	$P_{takeoff} = 86.67$	$P_{takeoff} = 76.35$
Max rate of climb requirement	$P_{ROCmax} = 66.96$	$P_{ROCmax} = 58.99$
Max velocity requirement	$P_{v_{\infty}max} = 38.61$	$P_{v_{\infty}max} = 34.01$

6.0 Conclusion

The motivation for this aircraft design was to aide in search and rescue operations where no passengers or crew had to be on the aircraft. The initial weight estimates were based on the PA-28-180 because it is a popular single-engine aircraft with similar performance to what we were looking to get out of our aircraft. The cruise speed, stall speed, takeoff distance, and landing distance initial estimates were all based off the PA-28-180. Some basic calculations based on the search area we wish to cover on a single flight we determined the appropriate range of 1000 nautical miles. A cruise altitude was set to a low value of 3000ft so that the reconnaissance equipment has a better resolution of the ground. The final design specifications obtained throughout the design iteration process can be seen in the appendix. Using these first estimations we calculated the wing loading, the drag coefficients, thrust to weight ratio's and lastly the power required based on the flight parameters. This process was iterated five times to reach the final values used in our final design seen in appendix.

7.0 Appendix

Weight (R1)

Step 1

Range	6076000	ft
npt	0.85	
c	1.4165E-07	1/ft
L/Dmax	14.2	
W3/W2	0.931176835	

Step 2

W1/W0	0.97	
W2/W1	0.985	
W4/W5	1	
W5/W4	0.995	
W5/W0	0.885244442	

Step 3

Wf/W0	0.114755558	
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Step 3

Wpayload	334.25	lb
We/Wo	0.62	
Wo	1110.14	lb

W/S (R2)

Step 1

rho	0.002377	lbs/ft3
Clmax, landing	2.93	

Vstall	84.3905	fps
W/S	24.77704403	lb/ft2

Step 2

Landing distance	1700	ft
L/W	1.2	
g	32.2	ft/s2
R	1673.059844	ft
mu	0.4	

thetaA	3	degrees
hf	2.292870369	ft

Sa	910.3062615	ft
Sf	87.56118678	ft
Sfr	291.147225	ft
Sg	365.6258912	ft
Total Landing Distance	1654.640564	ft

W/S (R3)

g	32.17	f/s^2
rho	0.002377	slugs/ft^3
Takeoff clearance	50	ft
Takeoff distance	775	ft
Clmax, take off	2.05	
Sg	190.8265797	*1/(T/W).7Vlo

R	1540.796555	ft
θob	0.255451721	rad
Sa	389.3323201	ft

(T/W)vlo	0.494795363	
----------	-------------	--

P(takeoff distance)	76.34942807	hp
---------------------	-------------	----

ROC max	12	ft/s
Cfe	0.0043	
Swet/s	4	
Cdo	0.017	
K	0.072931374	
eo	0.6	
AR	7.274179536	
P(ROC)	58.99210685	hp
Vmax	219.4	fps
Cruise alt	3000	ft
density at 2000ft	0.002409	
w2	1060.682642	
Wmc	1004.28352	
Wmc/s	22.41446265	
T/wmc	0.072168861	
P(vmax)	34.01427327	hp

Design Requirements (R1, R2, R3)	
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R1 Payload	
------------	--

Canon EOS 5D Mark IV DSLR Camera	2
Infrared (IR) Camera FLIR Boson 640	0.15
Multispectral Cameras MicaSense RedEdge-MX	0.8
LiDAR	1.3
Synthetic Aperture Radar (SAR) ceye X3 (Miniature SAR Satellite)	110
uni-pac ii x2	220

R2	
----	--

Landing Distance [ft]	2500
V Stall [Mi/h]	50

R3	
----	--

Takeoff Distance [ft]	2500
Rate of Climb Max [ft/min]	650
V Max at Midcruise Weight [Mi/h]	100

Wing Deatils	
--------------	--

AR (Aspect ratio, from part 1)	7.27
S (Wing Area, from part 1)	44.81 [ft^2]
Taper Ratio	0.5
b (wing span)	18.05 [ft]
Cr (Root chord)	3.31 [ft]
Ct (Tip Chord)	1.65 [ft]
C_bar	2.57

Fuselage Details		
Plane Length	20.00	[ft]
W Engine i = 1	222.60	[lb]
Engine Length	1.83	[ft]
W Avionics Bay i = 2	114.25	[lb]
W Payload i = 3	220.00	[lb]
X _{i=1}	19.08	[ft]
X _{i=2}	1.00	[ft]
X _{i=3}	6.00	[ft]
X _{ac} *	10.20	[ft]
bf (Fuselage Width)	3	[ft]

CG of Aircraft		
C _{fuselage}	3.03	
S Exposed	35.29	[ft ²]
W Wing i = 4	88.23	[lb]
X _{i = 4}	10.59	
CG of Aircraft	10.26	

Tail Details	
V _{HT}	0.7
L _{HT}	9.74
S _{HT}	8.29
Aspect Ratio	4.00
b _{HT}	5.76
V _{VT}	0.04
L _{VT}	9.74
S _{VT}	3.32
Aspect Ratio	1.50
h _{VT}	2.23

Stability	
-----------	--

a	4.309
d(C_M,CG)/dα	-4.747

W1 - Constant Weight		[lbs]
Wing Weight	88.23	[lbs]
Horizontal Tail Weight	16.57	[lbs]
Verticle Tail Weight	6.64	[lbs]
Fuselage Weight	76.05	[lbs]
S_Wet	54.32	[ft^2]
S_A (Elliptical Dome)	7.07	[ft^2]
S_C (Circular Cylinder)	36.00	[ft^2]
S_C (Trapazoidal-Cone)	11.25	[ft^2]
Landing Gear Weight	63.28	[lbs]
Dry Engine Weight (Wo)	159.00	[lbs]
Installed Engine Weight	222.60	[lbs]
All Else Empty Weight	15.90	[lbs]

Total Empty Weight	648.26
Revised Take off Weight	1109.91

Takeoff Distance	
------------------	--

Takeoff Distance	580.1588999
Sg	190.8265797
Sa	389.3323201

Landing Distance	
------------------	--

Landing Distance	1654.640564
Sa	910.3062615
Sf	87.56118678
Sfr	291.147225

Sg

365.6258912

[OBJ]

References

- [1] The National Centre for Missing Persons and Unidentified Remains (NCMPUR), "2022 Fast Fact Sheet," The Government of Canada, 2023.
- [2] CBC News, "By the numbers," 12 05 2009. [Online]. Available: <https://www.cbc.ca/news/canada/by-the-numbers-1.801937>. [Accessed 14 04 2024].
- [3] US Air Force, "MQ-1B Predator," 09 2015. [Online]. Available: <https://www.af.mil/About-Us/Fact-Sheets/Display/article/104469/mq-1b-predator/>. [Accessed 14 04 2024].
- [4] Global Air, "PIPER CHEROKEE PA-28-180," [Online]. Available: <https://www.globalair.com/aircraft-for-sale/specifications?specid=101>. [Accessed 28 03 2024].
- [5] J. D. Anderson, AIRCRAFT PERFORMANCE AND DESIGN, Tata McGraw-Hill, 2012.
- [6] "THE VIKING 90," VIKING AIRCRAFT ENGINES, [Online]. Available: <https://www.vikingaircraftengines.com/engine-options-1>.
- [7] "NACA 65(2)-415," Airfoil Tools, [Online]. Available: <http://airfoiltools.com/airfoil/details?airfoil=naca652415a05-il>.
- [8] C. O'Neil, "AeroFluids Answers," 2020. [Online]. Available: <https://charles-oneill.com/blog/cherokee-tapered-wing-float/>.
- [9] S. F. R. Elmer A. Horton, "TECHNICAL NOTE 2149 - INVESTIGATION OF BOUNDARY-LAYER CONTROL TO IMPROVE THE LIFT AND DRAG CHARACTERISTICS OF THE NACA 652~415 AIRFOIL SECTION WITH DOUBLE SLOTTED AND PLAIN FLAPS," NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS, 1950. [Online]. Available: <https://apps.dtic.mil/sti/pdfs/ADA380475.pdf>.